CS 556: Mathematics for Machine Learning
Assignment-11

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Question 1 (24 points)

Mopeds (small motorcycles with an engine capacity below 50 cm^3) are very popular in Europe because of their mobility, ease of operation, and low cost. The article "Procedure to Verify the Maximum Speed of Automatic Transmission Mopeds in Periodic Motor Vehicle Inspections" (J. of Automobile Engr., 2008: 1615–1623) described a rolling bench test for determining maximum vehicle speed. A normal distribution with mean value 46.8 km/h and standard deviation 1.75 km/h is postulated. Consider randomly selecting a single such moped.

(a) What is the probability that maximum speed is at most 50 km/h?

(b) What is the probability that maximum speed is at least 48 km/h?

(c) What is the probability that maximum speed differs from the mean value by at most 1.5 standard deviations?

Solution

Let X denote the moped's max speed: $X \sim \mathcal{N}(46.8,\, 1.75^2)$

(a)

$$Z = \frac{50 - 46.8}{1.75} \approx 1.83$$

$$P(X \leq 50) = P(Z \leq 1.83) \approx 0.9664$$

(b)

$$Z = \frac{48 - 46.8}{1.75} \approx 0.69$$

$$P(X \geq 48) = 1 - P(Z < 0.69) \approx 1 - 0.7549 = 0.2451$$

(c)

$$P\big(|X - 46.8| \leq 1.5 \times 1.75\big) = P(44.175 \leq X \leq 49.425)$$

$$Z_{\text{lower}} = -1.5, \quad Z_{\text{upper}} = 1.5$$

$$P(-1.5 < Z < 1.5) = 0.8664 - 0.0668 = 0.7996$$

Question 2 (38 points)

Suppose the diameter at breast height (in.) of trees of a certain type is normally distributed with mean 8.8 and standard deviation 2.8, as suggested in the article "Simulating a Harvester-Forwarder Softwood Thinning" (Forest Products J., May 1997: 36–41).

(a) What is the probability that the diameter of a randomly selected tree will be at least 10 in? Will exceed 10 in?

(b) What is the probability that the diameter of a randomly selected tree will exceed 20 in?

(c) What is the probability that the diameter of a randomly selected tree will be between 5 and 10 in?

(d) What value c is such that the interval $(8.8 - c,\ 8.8 + c)$ includes 98% of all diameter values?

(e) If four trees are independently selected, what is the probability that at least one has a diameter exceeding 10 in?

Solution

Let $X \sim \mathcal{N}(8.8,\, 2.8^2)$

(a)

$$Z = \frac{10 - 8.8}{2.8} = 0.43$$

$$P(X \geq 10) = 1 - P(Z < 0.43) \approx 1 - 0.6664 = 0.3336$$

(b)

$$Z = \frac{20 - 8.8}{2.8} = 4.0$$

$$P(X > 20) \approx 0$$

(c)

$$Z_1 = \frac{5 - 8.8}{2.8} = -1.36, \qquad Z_2 = \frac{10 - 8.8}{2.8} = 0.43$$

$$P(5 < X < 10) = P(-1.36 < Z < 0.43) = 0.6664 - 0.0869 = 0.5795$$

(d)

$$c = 2.33 \times 2.8 = 6.52$$

(e)

$$p = 0.3336$$

$$\text{All below 10: } (1 - p)^4 = 0.6664^4 \approx 0.1976$$

$$\text{At least one above 10: } 1 - 0.1976 = 0.8024$$

Question 3 (38 points)

Consider babies born in the "normal" range of 37–43 weeks gestational age. Extensive data supports the assumption that for such babies born in the United States, birth weight is normally distributed with mean 3432 g and standard deviation 482 g.

(a) What is the probability that the birth weight of a randomly selected baby of this type **exceeds** 4000 g? Is **between** 3000 and 4000 g?

(b) What is the probability that the birth weight of a randomly selected baby of this type is either less than 2000 g or greater than 5000 g?

(c) What is the probability that the birth weight of a randomly selected baby of this type exceeds 7 lb?

(d) How would you characterize the most extreme 0.1% of all birth weights?

(e) If X is a random variable with a normal distribution and a is a numerical constant ($a \neq 0$), then $Y = aX$ also has a normal distribution. Use this to determine the distribution of birth weight expressed in pounds (shape, mean, and standard deviation), and then recalculate the probability from part (c). How does this compare to your previous answer?

Solution

Let $X \sim \mathcal{N}(3432,\, 482^2)$

(a)

$$Z = \frac{4000 - 3432}{482} = 1.18$$

$$P(X > 4000) = 1 - P(Z < 1.18) = 1 - 0.8810 = 0.119$$

For $3000 < X < 4000$:

$$Z_1 = \frac{3000 - 3432}{482} = -0.90, \quad Z_2 = 1.18$$

$$P(3000 < X < 4000) = 0.8810 - 0.1855 = 0.6955$$

(b)

$$Z_{\text{low}} = \frac{2000 - 3432}{482} = -2.97, \quad Z_{\text{high}} = \frac{5000 - 3432}{482} = 3.25$$

$$P(X < 2000) + P(X > 5000) \approx 0.0015 + 0.0006 = 0.0021$$

(c) Convert 7 lb to grams: $7 \times 453.59 = 3175$ g.

$$Z = \frac{3175 - 3432}{482} = -0.53$$

$$P(X > 3175) = 1 - P(Z < -0.53) = 1 - 0.2970 = 0.703$$

(d) $z = 3.09$

Upper cutoff: $3432 + 3.09 \times 482 = 4920\,\text{g}$, Lower cutoff: $3432 - 3.09 \times 482 = 1944\,\text{g}$

(e) Convert to pounds: $Y = X/453.59$

$$Y \sim \mathcal{N}(7.57,\, 1.06^2)$$

For $Y > 7$:

$$Z = \frac{7 - 7.57}{1.06} = -0.54$$

$$P(Y > 7) = 1 - P(Z < -0.54) = 1 - 0.2946 = 0.705$$